

Kudankulam Nuclear Power Project Unit-5 Milestone

Source:ET

The **Nuclear Power Corporation of India Limited (NPCIL)** successfully completed the installation of the **Reactor Pressure Vessel (RPV)** at **Unit-5 of the Kudankulam Nuclear Power Project (KKNPP)** in Tamil Nadu, marking a significant milestone in India's clean energy expansion with Russian collaboration.

- The development comes after the **Atomic Energy Regulatory Board (AERB)** granted clearance for major equipment erection and the installation of key nuclear island components.
- **Indo-Russian Collaboration:** The installation was executed with technical support from **Rosatom** (Russia's state nuclear energy corporation) and its engineering division, Atomstroyexport (ASE). This partnership operates under a historic **Intergovernmental Agreement initially signed in 1988**.
- **Reactor Pressure Vessel (RPV):** The RPV is a massive steel component weighing approximately 320 tonnes. Often referred to as the "heart" of a nuclear power plant, it houses the reactor core where **nuclear fission** takes place to generate energy.
- **Installation Technique:** The RPV was installed using the highly precise **"open-top" method**. In this technique, heavy equipment is lowered directly into the reactor building using a high-capacity crane before the containment dome is erected and closed.
- **Reactor Technology:** KKNPP is being developed in stages, utilizing Russian-designed **VVER-1000** (Water-Water Energetic Reactor) technology. These are a specific class of **Pressurized Water Reactors (PWR)** that use light water as both a coolant and a neutron moderator.
- **Current Operational Status:** The currently operational units (Units 1 and 2) have already generated over 127 billion kWh of electricity.
 - This has significantly reduced carbon emissions (avoiding over 112 million tonnes of CO₂), minimized coal dependence, and advanced India's sustainable development goals.
- **Significance for Energy Security:** Once all six planned units of KKNPP become operational, the Kudankulam complex will achieve a total installed capacity of **6,000 MWe (Mega Watt electrical)**.
 - The project will emerge as a major nuclear power hub, **supporting India's goal of achieving 100 GW of nuclear capacity by 2047 under the Nuclear Energy Mission for Viksit Bharat**.

Read more: Strengthening India's Nuclear Energy Ecosystem

US - Iran Sign Islamabad MoU and its Implications

For Prelims: G7 Summit , Operation Epic Fury , Strait of Hormuz , International Atomic Energy Agency , Cape of Good Hope , Gulf Cooperation Council , Hezbollah

For Mains: Changing geopolitical dynamics in West Asia and their implications for global stability, Strategic significance of the Strait of Hormuz in global energy security.

Source: IE

Why in News?

A major breakthrough in **West Asian geopolitics** occurred as the US and Iran signed a landmark 14-point peace agreement, known as the **Islamabad Memorandum of Understanding (MoU)**.

- The MoU, signed on the sidelines of the **G7 Summit in France** , mediated by Pakistan, halts the intense direct and proxy military hostilities of **Operation Epic Fury** that erupted in February 2026, triggers the immediate lifting of the US naval blockade on Iranian ports, and opens the strategically vital **Strait of Hormuz** to unrestricted international maritime trade.

Summary

- The Islamabad MoU marks a major US-Iran diplomatic breakthrough, ending hostilities, reopening the Strait of Hormuz, initiating uranium downblending under IAEA supervision, and creating a framework for long-term negotiations on regional security and sanctions relief.
- The agreement is expected to ease global energy market pressures, reshape West Asian geopolitics, revive key connectivity projects such as Chabahar Port and INSTC, and enhance India's energy security and economic interests.

What is the Scope of the US - Iran Islamabad MoU?

- **Temporary Ceasefire Framework:** The Islamabad MoU serves as an interim diplomatic framework rather than a permanent peace treaty, providing for an immediate cessation of hostilities and initiating a 60-day negotiation process between the US and Iran on nuclear proliferation, regional security, and sanctions relief.
- The final comprehensive agreement is proposed to be formalized through a binding resolution of the **UN Security Council**.
- **The Nuclear Downblending Mandate:** The agreement only requires Iran to downblend its highly enriched uranium (HEU) under International Atomic Energy Agency (IAEA) supervision, while key verification mechanisms, timelines, and technical details remain subject to a 60-day

negotiation process.

- **Reciprocal Commitments:**

- The preliminary accord requires the US and its international partners to formulate a **USD 300 billion financial package** to fund Iran's post-war economic recovery.
- **US:** Committed to lifting its naval blockade, easing restrictions on Iranian oil exports, opening talks on broader sanctions relief.
- **Iran:** Committed to halting nuclear weapons development, ensuring the unhindered flow of commercial **shipping through the Persian Gulf** , and **engaging in final settlement negotiations**.
- **Reopening of the Strait of Hormuz:** Iran has agreed to grant **absolute toll-free transit** to all commercial vessels through the Strait of Hormuz for the duration of the 60-day negotiation window.
- The 14-point text dictates that maritime traffic through the channel must be restored to its full, pre-conflict capacity within 30 days of signing.

What is "Downblending"?

- **The Process:** Downblending is the nuclear process of mixing **Highly Enriched Uranium (HEU)** which has a high concentration of the Uranium-235 isotope with natural or depleted uranium.
- This process chemically dilutes the stockpile into **Low-Enriched Uranium (LEU)** , which is suitable for civilian power generation but not for weapons.
- **Strategic Significance:** Prior to the agreement, Iran was enriching uranium up to 60% purity, perilously close to the 90% weapons-grade threshold.
- By agreeing to downblend this stockpile , Iran functionally extends its **"breakout time"** (the time needed to build a nuclear bomb), thereby addressing core Western and regional security concerns.

[Click Here to Read: The Escalating Crisis in West Asia](#)

What are the Implications of the Islamabad MoU?

Global Economic and Energy Implications

- **Cooling of Global Energy Markets:** The immediate mandate to reopen the **Strait of Hormuz** has instantly erased the "geopolitical risk premium" on oil.
 - Global benchmarks like **Brent crude** corrected sharply, dropping by over 2%, signaling immediate relief for energy-hungry economies.
 - This eliminates the need for expensive maritime detours (such as rerouting around the **Cape of Good Hope**), drastically slashing exorbitant freight costs and **prohibitive war-risk insurance premiums for global shipping**.
- **Iran's Economic Resurgence:** The unfreezing of restricted assets, waivers on crude oil exports, and the proposed **USD 300 billion reconstruction fund** will reintegrate Iran into the global financial system, ending years of crippling economic isolation.

Regional Implications for West Asia

- **Strategic Isolation of Israel:** The MoU represents a massive diplomatic and strategic defeat for Israel. By mandating a permanent end to military operations on all fronts (including Lebanon), the agreement effectively shields **Hezbollah from destruction**.
 - In defiance, the **Israeli Defense Forces (IDF)** have vowed to continue operations in southern Lebanon, risking a severe diplomatic rupture with Washington.
- **The Gulf Security Dilemma:** For **Gulf Cooperation Council (GCC)** states like Saudi Arabia and the UAE, the MoU is a double-edged sword.
 - While they urgently required the reopening of the Strait of Hormuz for their own oil exports, they view the deal with deep apprehension due to fears that **Iran's financial windfall will fund its advanced ballistic missile program and regional proxy networks**.
- **Lebanon's Subordinated Sovereignty:** Tying the cessation of hostilities in Lebanon to the US-Iran "grand bargain" threatens to freeze Lebanon's political status quo.
 - This risks subordinating genuine Lebanese state-building to broader regional geopolitics, leaving armed non-state actors entrenched in power.

Geopolitical and Strategic Implications

- **Validation of the "Axis of Resistance":** By preserving key elements of its nuclear and ballistic missile capabilities, Iran has portrayed the agreement as a diplomatic victory, emerging both structurally validated and regionally emboldened.
- **US Paradigm Shift:** The US acceptance of Iran retaining a limited ballistic missile capability marks a **departure from the "maximum pressure" doctrine**, reflecting a more pragmatic foreign policy focused on regional stability and domestic economic priorities rather than prolonged Middle Eastern conflicts.
- **The Rise of Alternative Mediators:** Pakistan's successful mediation, supported by Qatar and Oman, highlights the growing role of regional middle powers in resolving major geopolitical conflicts.

What are the Implications of Islamabad MoU for India?

- **Macroeconomic Relief:** India imports roughly 85% of its crude oil requirements. The sharp drop in global crude prices will directly **compress India's import bill, rein in the current account deficit , and tame fuel-led domestic inflation**.
- Reduced transit times and slashed logistics overheads will immediately improve the global competitiveness of heavily crude-dependent Indian export sectors, such as **plastics, chemicals, textiles, and engineering goods** destined for Europe and West Africa.
- **Protection of the Indian Maritime Workforce:** **Accountable for nearly one-sixth of global seafarers, Indian maritime workers face acute risks from geopolitical conflicts, as exposed by the fatal MT Settebello strike.**
- **The halt in military attacks mandated by the Islamabad MoU can secure stranded Indian mariners.**
- **Revival of Connectivity Projects:** The lifting of the naval blockade on Iranian coastal waters clears the path for the unhindered operationalization of the India-backed **Chabahar Port** and accelerates the viability of the **International North-South Transport Corridor (INSTC)** , granting India vital, secure access to Central Asia and Russia.
- While the deal completely revitalizes the INSTC, it inadvertently stalls the **India-Middle East-Europe**

Economic Corridor (IMEC) due to Israel's continued strategic isolation and unilateral operations in Lebanon.

- **Diplomatic Tightrope:** While economically beneficial, the MoU complicates New Delhi's strategic balancing act.
- India will need to carefully recalibrate its deep strategic partnership with a **defiant Israel while simultaneously capitalizing on the newly opened economic avenues with an emboldened Iran.**
- **Impact of Pakistan's Role as the Mediator:** Acting as a mediator has strengthened Pakistan-Iran bilateral ties. This potentially clears the path for long-stalled cross-border energy projects (e.g., the [Iran-Pakistan Gas Pipeline](#)) free from the immediate threat of secondary US sanctions, shifting the energy dynamics in India's immediate neighborhood.

INDIA'S \$15 BILLION ENERGY 'WIN' & REGIONAL RISKS

THE ECONOMIC RELIEF



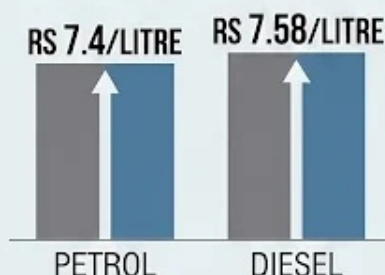
RELIEF from energy crisis for 400M middle-class families.

REDUCED IMPORT BILL shrinks current account deficit, eases pressure on Rupee.

\$15 BILLION

'WIN' for India if commitments honoured.

UNCERTAIN CONSUMER BENEFIT

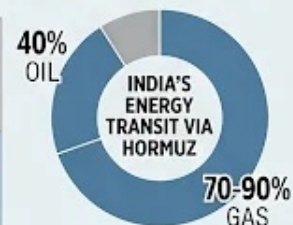


OMCs face **SIZEABLE COST-REVENUE GAPS** despite price hikes.



Lower prices uncertain; delayed Hormuz re-opening could keep them up.

HORMUZ CHOKEPOINT VULNERABILITY



OVER-DEPENDENCE remains a concern despite diversification. Re-start of conflict is a risk.



REGIONAL STABILITY & US-IRAN PEACE ESSENTIAL



SUSTAINABLE LONG-TERM PEACE CRITICAL for India's energy needs.



TARGETED by Iranian strikes, damaging energy architecture.

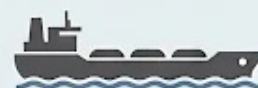
SANCTIONS RELIEF & FUTURE IMPORTS



US AGREED to lift sanctions on Iranian oil & petrochemicals.



POTENTIAL for renewed trade; India bought **300,000 BARRELS/DAY** from Tehran pre-sanctions.



Energy sales are ~**60%** of Iran's total revenue.

What Measures are Needed to Ensure Durable Peace in West Asia?

- **Establishing Comprehensive Verification Frameworks:** For the interim MoU to transition into a durable peace treaty during the upcoming Swiss round of negotiations, a verifiable, **IAEA-backed monitoring mechanism must be established** to oversee Iran's nuclear downblending and enrichment pauses.
- **Clarifying Global Chokepoint Jurisprudence:** The international community, under the **UN and the International Maritime Organization (IMO)**, must structurally decouple freedom of navigation in the Strait of Hormuz from regional political bargains, ensuring that the waterway cannot be legally or militarily leveraged as a geopolitical weapon after the 60-day toll-free period expires.
- **De-escalating the Israel-Lebanon Flashpoint:** Neutral international intermediaries must work to integrate Israel's immediate northern security anxieties into the core text of the upcoming Swiss negotiations.
 - Failing to address these concerns will allow unilateral military operations in southern Lebanon to completely shatter the fragile West Asian peace.

Conclusion

The Islamabad MoU represents a triumph of economic and political pragmatism over protracted conflict. While it provides immediate relief to global energy markets and boosts India's macroeconomic and trade landscape, its durability remains contingent on the upcoming 60-day negotiation window. Only by transforming this tactical pause into a comprehensive, inclusive regional security framework can West Asia transition from a state of managed crisis to permanent stability.

Drishti Mains Question:

Q. How does the Islamabad MoU alter the geopolitical balance of power in West Asia? Analyze its implications for regional actors.

Frequently Asked Questions (FAQs)

1. What is the Islamabad MoU between the US and Iran?

The Islamabad MoU is a 14-point interim peace agreement that halts hostilities, initiates nuclear negotiations, and facilitates the reopening of the Strait of Hormuz.

2. What is meant by downblending of uranium?

Downblending is the process of diluting **Highly Enriched Uranium (HEU)** into **Low-Enriched Uranium (LEU)**, making it unsuitable for nuclear weapons and suitable for civilian purposes.

3. Why is the Strait of Hormuz strategically important?

It is one of the world's most critical energy chokepoints, through which a substantial share of global oil and LNG trade passes.

4. How does the Islamabad MoU benefit India?

The agreement lowers crude oil prices, reduces shipping costs, strengthens the viability of **Chabahar Port** and **INSTC**, and improves India's energy security.

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Which one of the following countries of South-West Asia does not open out to the Mediterranean Sea? (2015)

- (a) Syria
- (b) Jordan
- (c) Lebanon
- (d) Israel

Ans: B

Q. The term “two-state solution” is sometimes mentioned in the news in the context of the affairs of (2018)

- (a) China
- (b) Israel
- (c) Iraq
- (d) Yemen

Ans: B

Mains

Q. “India’s relations with Israel have, of late, acquired a depth and diversity, which cannot be rolled back.” Discuss. (2018)

Indian Ocean Rim Association (IORA)

Source: TH

The **Indian Ocean Rim Association (IORA)** is examining **Canada’s application to become a dialogue partner** at the 28th Committee of Senior Officials meeting in New Delhi, reflecting the organisation’s expanding maritime engagement.

- Canada is being considered due to its **extensive maritime expertise, including large coastal zones and experience in maritime safety, security, connectivity**, and ocean governance, which could benefit IORA member states.

Indian Ocean Rim Association (IORA)

- **About:** The Indian Ocean Rim Association (IORA) is **an intergovernmental organisation of 23 Member States and 12 Dialogue Partners**, promoting regional cooperation in the Indian Ocean region.
- **Origin of Idea:** The concept was **first articulated by South African President Nelson Mandela during his 1995 visit to India**, emphasising natural historical and geographical linkages of Indian Ocean states.
- **Evolution:** This idea led to the **Indian Ocean Rim Initiative in 1995**, which eventually resulted in the formation of IORA in 1997.
- **Charter:** The IORA Charter serves as the foundational document guiding cooperation based on multilateralism, consensus, and non-intrusive engagement among member states.
- **Priority Areas:** IORA focuses on **six priority areas** and **two cross-cutting issues** to promote sustainable regional cooperation and development in the Indian Ocean region.
- **Institutional Mechanism:** The organisation functions through institutional structures including the **Council of Ministers (COM), Committee of Senior Officials (CSO), TROIKA, and the IORA Secretariat**.
- **Member States:** IORA includes key Indian Ocean littoral states such as **India, Australia, Indonesia, Malaysia, South Africa**, Sri Lanka, Oman, Kenya, Tanzania, Mozambique, Madagascar, Mauritius, Seychelles, Somalia, Comoros, Maldives, France, UAE, Iran, Bangladesh, Thailand, Singapore, and Yemen, among others.
- **Dialogue Partners:** IORA's dialogue partners include major global and regional actors such as the **United States, United Kingdom, China, Japan**, European Union, Russia, Germany, Italy, Republic of Korea, Türkiye, Saudi Arabia, and Egypt, reflecting its widening geopolitical engagement.
- **Global Linkages (Observer Status):** IORA holds observer status in key global institutions including the **United Nations General Assembly (UNGA)**, **African Union (AU)**, and **UNCTAD**, enabling deeper international cooperation and representation of Indian Ocean interests in global governance.



Kishau Multipurpose Dam Project

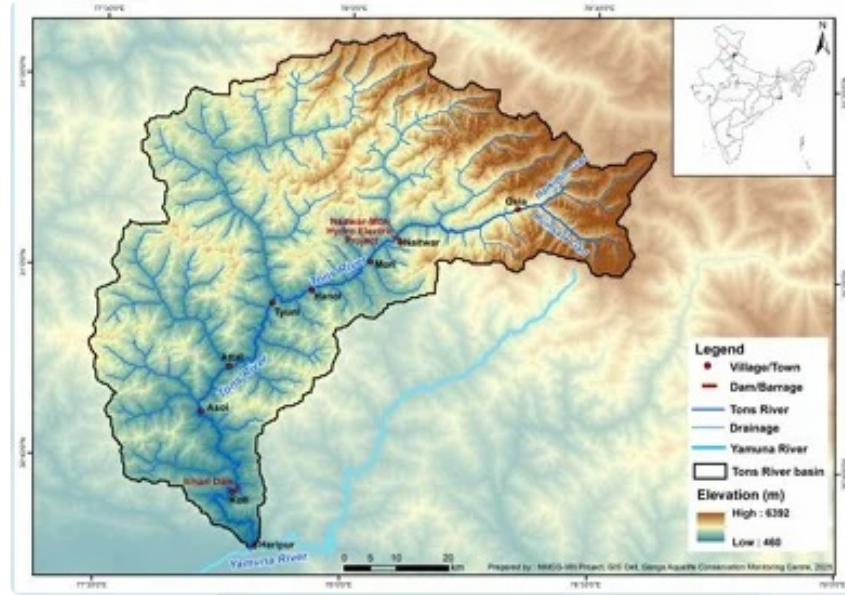
Source: PIB

A consensus has been reached among six states for the **Kishau Multipurpose Dam Project on the Yamuna basin** , enabling inter-state water sharing and financing arrangements, marking a major step in river rejuvenation and cooperative federalism.

- **About:** The Kishau Multipurpose Project is a long-delayed **inter-state hydropower and water storage project on the Tons River (a major tributary of the Yamuna)** , aimed at irrigation, drinking water supply, and power generation.
- **States Agreement:** **Himachal Pradesh, Uttarakhand, Delhi, Uttar Pradesh, Haryana, and Rajasthan** agreed to sign a Memorandum of Understanding (MoU) for implementation of the project.
- **Funding Structure:** The **Central Government will bear 90% of the water component cost as Central Assistance** , while the remaining **10% will be shared by six participating states** .
- **Water Allocation Formula:** It was agreed that **Himachal Pradesh's share of water will be allocated to Delhi and Rajasthan**, in return for sharing the cost of Himachal Pradesh's share in the power component of the project.
- **Significance:** The project is expected to ensure **cleaner and increased flow in the Yamuna** , supporting drinking water supply to Delhi, irrigation needs, and hydropower generation.

Tons River

- **Origin & course:** It is the longest **Himalayan tributary** of the Yamuna, originating from the **Bandarpunch massif (6,315 m)** and flows about **150 km** through Uttarkashi and Dehradun (Uttarakhand) and Sirmaur (Himachal Pradesh), finally meeting the Yamuna near **Kalsi (Dehradun)** .
- **Physiography & Tributaries:** The river flows through **rugged Himalayan terrain with steep slopes, V-shaped valleys and glaciated catchments** , and is fed by tributaries such as **Pabar, Sainj, Nira, Har ki Dun Gad, and Ruinsara Gad** .
- **Significance:** It forms a **natural boundary between Uttarakhand and Himachal Pradesh** in the Jaunsar region, highlighting its geopolitical and hydrological importance.



[Read More: Siang Upper Multipurpose Project](#)